



Analysis of herd manager daily routines and interrelations with production and health and economic key figures of dairy farms—A case study

Melanie Schären-Bannert,^{1,*†} Benno Waurich,² Adriana Wöckel,¹ Wolf Wippermann,¹ Christina Felgentreu,³ Franz Fröhlich,³ Julia Wittich,³ Fanny Rachidi,¹ Detlef May,³ Sven Dänicke,⁴ Hermann H. Swalve,² Alexander Starke,¹ and Peter Hufe³

¹Clinic for Ruminants and Swine, Faculty of Veterinary Medicine, University of Leipzig, 04103 Leipzig, Germany

²Institute of Agricultural and Nutritional Sciences, University of Halle, 06120 Halle-Wittenberg, Germany

³LVAT—Institute for Animal Breeding and Husbandry, 14550 Groß Kreutz, Germany

⁴Institute of Animal Nutrition, Friedrich-Loeffler-Institute (FLI), Federal Research Institute for Animal Health, 38116 Brunswick, Germany

ABSTRACT

Work organization is of paramount importance when it comes to creating efficient workflows to achieve high productivity in a complex operation such as a dairy farm. The person responsible for this task in mid- to large-scale dairy farms is the herd manager. We hypothesized a positive relationship between the invested hours of labor in herd management, the extent of controlling activity of herd managers, and the production and health status of the herd, as well as economic traits. A case study was conducted on 10 farms located in the state of Brandenburg with an average size of 592 German Holstein cows (229–1,221), an average milk production of 9,690 kg of marketed milk per cow and year, and 32.9% culling rate. In each farm, the person responsible for herd management was followed throughout a 3-wk period (mean: 13 d; 11–15, minimum–maximum [min-max]) by a project employee. Activities were continuously documented on a minute basis. An average of 954 entries (=documented activity, 537–1,326; min-max) and a total of 109 h (86–134 h) were recorded per farm and assigned to 13 categories during data processing. Herd data included key figures in the following areas: production level (kilogram of marketed milk per cow and year), milk quality (bulk tank and test-day data), stillbirth rates, and culling (culling and mortality rates). The herd managers invested 4% (1%–9%, min-max) in administration, office, and management, 17% (9%–23%) in communication, 11% (3%–19%) in animal documentation, 17% (11%–28%) in animal organization and movement, 9% (4%–16%) in animal controls, 9% (2%–16%) in animal treatment and health measures, 1%

(0%–3%) in feeding, 12% (4%–29%) in accompanying of external service providers (e.g., veterinary, hoof trimmer, inseminator), 5% (1%–11%) in assistance of employees, 5% (0%–30%) in shift takeover from employees, 5% (0%–9%) in process controlling, 2% (1%–5%) in other activities, and 3% (0%–11%) of the time was not assigned/not recorded. The 5 farms where herd managers invested more than average time in animal and process controlling (animal health, feeding, and other, $\geq 14\%$) lower SCC (288 thousand cells/mL vs. 205), and showed more favorable culling and replacement figures with a numerically lower culling rate (35.3% vs. 30.4%), interrelating with a higher amount of cow calvings (65.8% vs. 73.2%) and a higher lifetime production (30,400 kg vs. 36,698 kg), a lower mortality rate (9.7% vs. 5.9%), as well as less losses during the rearing period until 3 mo of age (3.5% vs. 8.3%). We concluded that a large heterogeneity between farms in the activities of their herd managers exists, and herd managers who spent more time in controlling to achieve better results concerning the health and longevity of their herd. The developed protocols and analysis routines should be applied in future studies, including a larger number of farms, and are a valuable toolbox for problem analysis and consultancy in practice.

Key words: work organization, workflow, controlling, monitoring

INTRODUCTION

Dairy production has evolved from traditional farming to modern and large-scale production systems in many parts of the world (Barkema et al., 2015; Britt et al., 2018). These systems imply the coordination and management of different complex workflows and the corresponding workforces (Hutt, 1993; Hesse et al., 2017; Mills et al., 2020). The person responsible for this task in mid- to large-scale dairy farms is the herd manager (New Zealand Government, 2024).

Received June 28, 2025.

Accepted September 23, 2025.

*Corresponding author: melanie.schaeren@uni-leipzig.de

†Current address: Unit Herd Health Management, Clinical Department for Farm Animals and Food System Science, University of Veterinary Medicine Vienna-Vetmeduni, 1210 Wien, Austria.

The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-25. Nonstandard abbreviations are available in the Notes.

Well-organized and efficient workflows allow the completion of tasks in an adequate manner and increase the quality of the result, as well as the working conditions for employees (Brass, 1985; Reijers et al., 2016; Rodriguez, 2021). Therefore, optimal process structuring/planning, management, and control are essential, also in dairy production systems (Hutt, 1993).

During the last decades different key figures and critical control points have been established to monitor dairy cow herd health, often implemented in software tools and programs for routine evaluation (de Kruif et al., 2006; Sanftleben et al., 2007; Green and Bradley, 2012; Noordhuizen, 2012). Also, the invested hours of labor in general (Schick et al., 2009; Pouloupoulou et al., 2018) and in specific areas (e.g., milking; Pitkäranta et al., 2019; Edwards et al., 2020; Prendergast et al., 2024; or calf care; Gleeson et al., 2008), as well as the efficiency of different routines and systems, have been subjected to investigation (Bewley et al., 2001; Hogan et al., 2022, 2023; König et al., 2023). Furthermore, studies have shown the level of education and expertise is a major risk factor for productivity and animal health in dairy production systems (Hutton et al., 1990; Yeamkong et al., 2010; Hansen and Greve, 2015; Sauer and Latacz-Lohmann, 2015; Lindena and Hess, 2022). However, protocols to describe and analyze farm workflows and invested hours in herd management and the distinct activity patterns of herd managers, as well as studies linking it to the herd health status and productivity, are lacking (PubMed Search, 27.7.2024, keywords: dairy, workflow analysis, herd management, farm management, herd health, productivity, labor, employee, work quality, work organization; keywords applied in different combinations; zero relevant hits). The control of production and working processes, the “controlling activity” of a herd manager, and its consequence for the health and production of the herd have rarely been reflected upon in the literature or investigated.

We hypothesized a positive relationship between the invested hours of labor in herd management, the extent of controlling activity of herd managers, and the production and health status of the herd, as well as economic traits. The aim of this case study was to develop respective protocols and to assess in a sample of dairy farms the hours invested in herd management, categorizing herd manager activities, and analyzing interrelations with the key figures for production, herd health, economics, and efficiency.

MATERIALS AND METHODS

The study was conducted within a European Innovation Partnership for Agricultural Productivity and Sustainability (EIP-AGRI project), implemented as a

multiactor operational group titled “Die Entwicklung des KUH-mehr-WERT Navigators” (German; “Developing the Cow Value Navigator;” Schären-Bannert et al., 2023). The EIP-AGRI is an EU framework that supports practice-oriented agricultural innovation through multi-actor collaboration.

Farms

The project involves 12 dairy farms located in the state of Brandenburg, being part of the test herd program of the breeder association RBB Rinderproduktion Berlin-Brandenburg GmbH, implying a high standard in animal production and health data documentation established on the farms (contract herds described in König and Swalve, 2024). Ten of these 12 farms volunteered for this specific data acquisition (Table 1). Two farms could not be included, as during the respective timeframe another data acquisition was conducted for which these 2 farms did not fulfill the criteria. The farms exhibited an average herd size of 592 German Holstein cows, an average milk production of 9,690 kg of marketed milk per cow and year, and 32.9% culling rate. All farms are characterized by a freestall confinement system for the milking herd with cubicles and, in most cases, dry cows on deep-bedded straw pack (except for farms P, Q, and Z, with the dry cows [partially] on pasture) and a TMR or partial (PMR, in case of robot milking) mixed ration feeding system. Flooring and cubicle type, milk systems, and other farm characteristics are given in Table 1. All farms ran an all-year calving system with minimal to no seasonal variation in the daily routines/work processes throughout the year.

Labor Hours and Herd Manager Activity Assessment

The daily workflows of the different positions on the farm were assessed in detail in interviews with the farm or herd manager, or both, using a standardized scheme (Figure 1) based on aspects described in Mothes et al. (1972) and Kleiber (1981). The activities within each position filled by an employee were assigned to 6 different areas: milking, housing management, feeding, reproduction, other husbandry tasks and treatments, and herd management, as described in Figure 1. All farms adhered to strong weekly routines, with overtime hours being very uncommon.

In contrast to all other positions, most farm and herd managers were not able to reproduce the sequence of daily activities of the herd manager and their respective duration reliably. Furthermore, the interviews revealed that the herd managers were often also occupied in the other areas (milking, feeding, reproduction, calves and youngstock, treatment, and control of animals).

Table 1. Description of project farms^{1,2}

Farm	O	P	Q	R	S	T	U	X	Y	Z
No. of cows	560	229	783	1,221	339	317	748	794	525	407
No. of youngstock/cow	1.02	0.76	0.82	0.88	0.92	0.76	0.74	0.53	0.75	0.89
Milk production (kg ECM/cow/yr)	10,038	9,746	10,330	9,782	9,089	8,943	9,235	10,549	10,001	9,527
Annual culling rate (%)	33.6	25.8	30.8	38.1	38.6	33.1	33.4	21.5	37.3	36.4
Feeding system	TMR	PMR	TMR	TMR	PMR	TMR	TMR	TMR	TMR	TMR
Flooring	Concrete	concrete	Concrete	Slatted	Concrete	Concrete	Concrete and rubber	Concrete	Concrete	Slatted
Cubicle surface/bedding type	Semideep bedded, lsw	Deep bedded, lsw, and rubber mattresses	Deep bedded, lsw	Rubber mattresses	Deep bedded, lsw	Deep bedded, straw	Deep bedded, lsw	Deep bedded, lsw	Deep bedded, lsw, and rubber mattresses	Rubber mattresses
Parlor type	2 × 12 units side-by-side	5 single AMS, 1 × 6 units herringbone	44 units external rotary	60 units external rotary	5 single AMS	20 units internal rotary	50 units external rotary	40 units external rotary	28 units internal rotary	2 × 12 units herringbone
Milkings/cow per d	3	*	2–3	2–3	*	2–3	2	2	2	2

¹Key figures from period July 2020 to June 2021.²ECM = annual culling rate; excluding sales for breeding purposes; PMR = partially mixed ration (concentrate fed at milking robot). Cubicle type: lsw = lime straw water mixture. *Data not available. Parlor type: AMS = automated milking system.

It was therefore decided to assess the activities of the herd managers by a real-time documentation adapted from Auernhammer (1976). The herd managers were each followed throughout a 3-wk period by a project employee (mean: 13 weekdays; 11–15, minimum–maximum [min-max]; 5-d wk, no weekends or public holidays included, days where herd managers were sick or not involved in the milk production process were not included). To minimize a possible observer bias, a focus was put on the discreteness of the observing person. Furthermore, the research question and hypothesis were not communicated to the herd managers involved in advance. Activities were continuously documented on a minute basis. An average of 954 entries (= documented activities, 537–1,326; min-max) and 109 h (excluding breaks and tasks in other areas than dairy production, 86–134 h) were recorded per farm. From the activities documented, a 3-level category system was developed (Table 2). All entries were then assigned to activities within the category system during data processing.

The % of the 12 main categories (Table 2) of the total number of hours of activities recorded was then calculated for each herd manager.

For the analysis of the farm workflows, the herd manager activities were then also assigned to the 6 areas: milking, housing management, feeding, reproduction, other husbandry tasks and treatments, and herd management, allowing the total hours specified for the herd manager position to be proportionally distributed to the respective areas.

Thereafter, the invested hours of labor were calculated for each area, and the following key figures were determined: hours of labor per cow and year (h/cow/yr) and minutes of labor per 100 kg of marketed ECM (min/100 kg of ECM). Furthermore, the average salary per labor hour (Euro/h) was calculated using the total costs for personnel indicated in the benchmark (see below) divided by the total hours of labor invested in the equivalent timeframe.

In an expert round discussion and considering literature from other production processes (Jones et al., 2001), we considered following main categories within our categorization system as controlling activity: animal health controls, feeding, and controlling of the production process (subcategories and details in Table 2). This categorization was then used for statistical comparison between groups (see subsection Statistical Analysis below).

Production and Herd Health Key Figures Assessment

In the scope of the overarching project, a benchmark containing 217 key economical as well as production figures was created on a quarterly basis between June

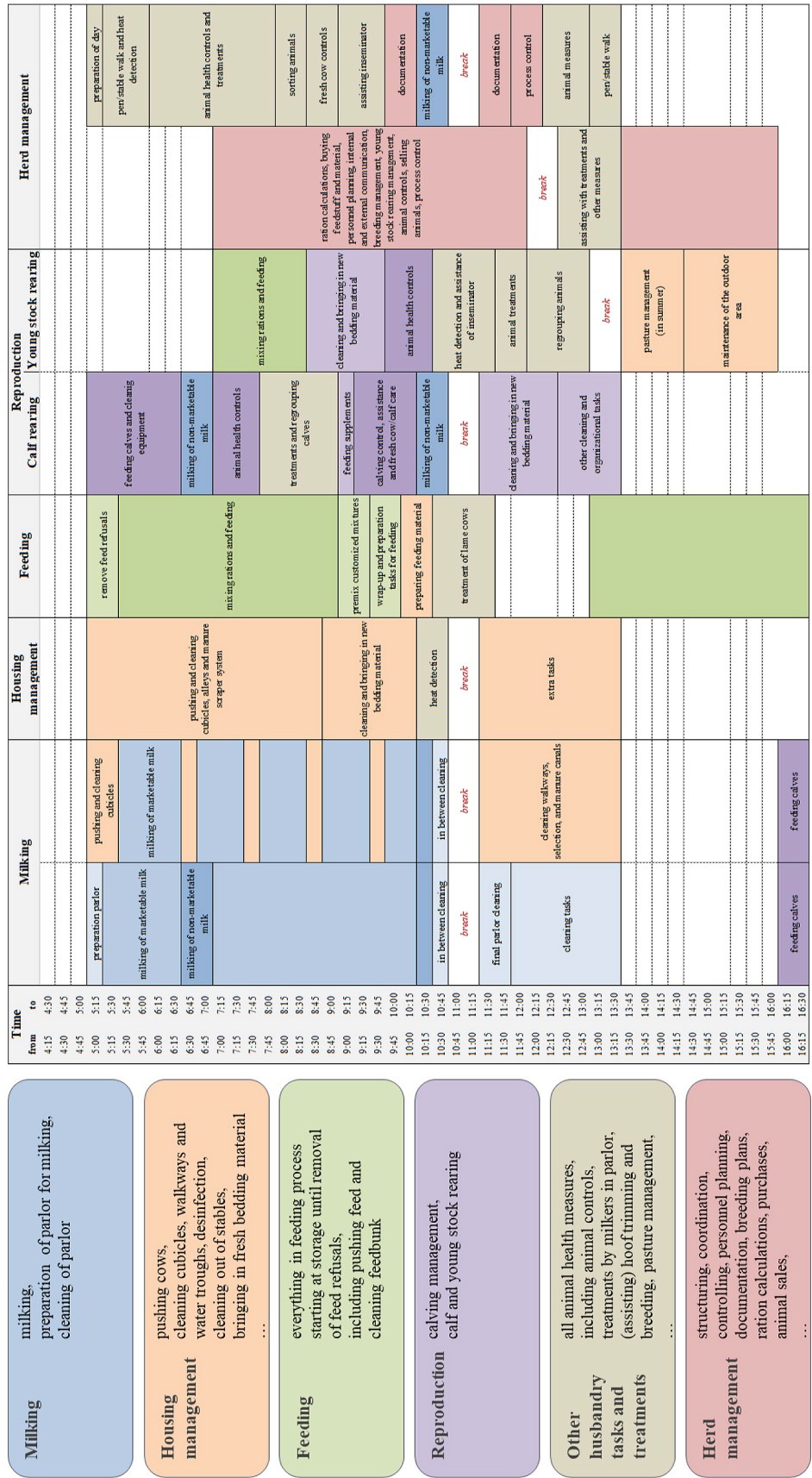


Figure 1. Example of daily farm workflow (farm X, 794 cows, cut at 1630 h for illustrational purposes). Each column represents a position filled by an employee. The occupations were assigned to 6 areas (described on the left) and colored accordingly. Total labor hours amounted to 36.5 h/cow/yr.

Table 2. List to describe and document herd manager activities divided into 12 main categories and 84 subcategories of the first level (left column) and 141 subcategories of the second level (right column)¹

Main category and first-level subcategory	Second-level subcategory
Administration, office, and management	
Meeting with supervisor	Meetings, making and correcting shift schedules
Personnel management	—
Scheduling own work	Purchasing products, mail, and billing
Business administration	—
Varia office management	Authorities, inseminator, nutritionist, hoof trimmer, technical company, veterinarian, livestock dealer
Communication	Employees in milk production, supervisor, colleagues in other farm branches
External	Treatments and medications, inseminations and breeding, animal identification, pen change, calvings, cullings
Internal	“
Animal documentation	—
Digital documentation in herd management program	For breeding, hoof trimming, test-day, veterinarian, animal health measure, pen change
Analog documentation	“
Reporting animals to the governmental database	“
Animal organization and movement	“
Creating animal list	“
Searching and marking animals	Pen change due to production stage, disease, or other reason
Enter animal for selection by selection gate	“
Separating single animal in group	—
Fixing animals in headlock in group	For sales for slaughter, breeding, or other
Moving and fixing animals in separate area outside pen	Changing or cleaning collars, insert ear tags, or all of these; changing or cleaning of transponder/RFID, or both
Moving single animal	Milk production drop and other sensor systems
Moving several animals	—
Transporting animals within farm	Checking sensor data, pen walks
Loading animals for transport	Single animal, mastitis group, fresh cows, sick cow pen, calving pen/surveillance, lactating groups, dry cows
Animal identification	Single animals (hutches), group pen, checking milk feeder automatic alarm list
Animal control	Single animals, group pen
Checking sensor alarm lists for health controls	“
Checking production and sensor data of single animal	Heifers, cows
Heat detection	Calves, youngstock, heifers, fresh cows, cows in milking parlor, cows or heifers during hoof trimming
Health controls cows (pen walks and health checks)	Calves, youngstock, heifers, cows
Health controls calves	—
Health controls youngstock	—
Health controls heifers	—
Animal (health) measures and treatments	Calves, youngstock, heifers, cows
Breeding	Heifers, cows
Treatment of sick animals	—
Dehorning	—
Vaccination	—
Drying off	—
Collecting milk samples	Calves, youngstock, heifers, cows
Blood sampling	—
Calving assistance	—
Hoof bath	Heifers, cows
Feeding	—
Ration calculation and adaption	—
Transmission of recipe to feed loading installation/mixer	—
Control of storage conditions and hygiene	—
Control of inventory	—
Feed sample collection	—
Control of feed refusals	—
Data analysis and interpretation	Loading protocols of feed mixer, silage sample analysis

Continued

Table 2 (Continued). List to describe and document herd manager activities divided into 12 main categories and 84 subcategories of the first level (left column) and 141 subcategories of the second level (right column)¹

Main category and first-level subcategory	Second-level subcategory
Accompanying external service providers	
Inseminator	Breeding, pregnancy checks
Nutritionist	—
Hoof trimmer	—
Test-day staff	—
Technical service provider/locksmith	—
Veterinarian	Examination and treatment, vaccination, dehorning, fertility/pregnancy checks
Livestock dealer	—
Assisting employees	—
Calving area	—
Milking	—
Pushing cows	—
Housing management	Cleaning out pens, cleaning cubicles, bringing in fresh bedding material, cleaning water troughs
Maintenance of technical equipment	—
Feeding	Cleaning feed bunk, pushing feed, filling and unloading feed mixer (feeding cows)
Calf and youngstock rearing	—
Shift takeover from employees	—
(same subcategories as in assistance of employees)	
Process control	
Data analysis	Cullings, calvings, test-day data, ADG calves and youngstock
Checking/organizing cows in wrong pens	—
Assisting in test-day recording	—
Weighing and measuring animals	Calves, youngstock, heifers
Milking and milk storage	Data analysis parlor, testing for antibiotics residues in bulk tank, checking storage conditions and hygiene, checking milking equipment condition and hygiene
	Calibration/control of calf feeder automat, control of drinking/feeding hygiene
Calf feeding	Manure scrapers/pumps/technique, water troughs, ventilation/climate, feed pushing robot
Control of technical equipment	Number of animals, consumables
Inventory	—
General control of cleanliness and order	—
Other activities	—
Break	—
Managing work clothing	—
Reading literature	—
Continuing education, training, external event	—
Tasks outside dairy production	—
Time off, private appointment	—

¹For illustrational purposes, the second level subcategories are listed in 1 table row and separated by commas. “ = same as in row above; — = no second-level subcategory.

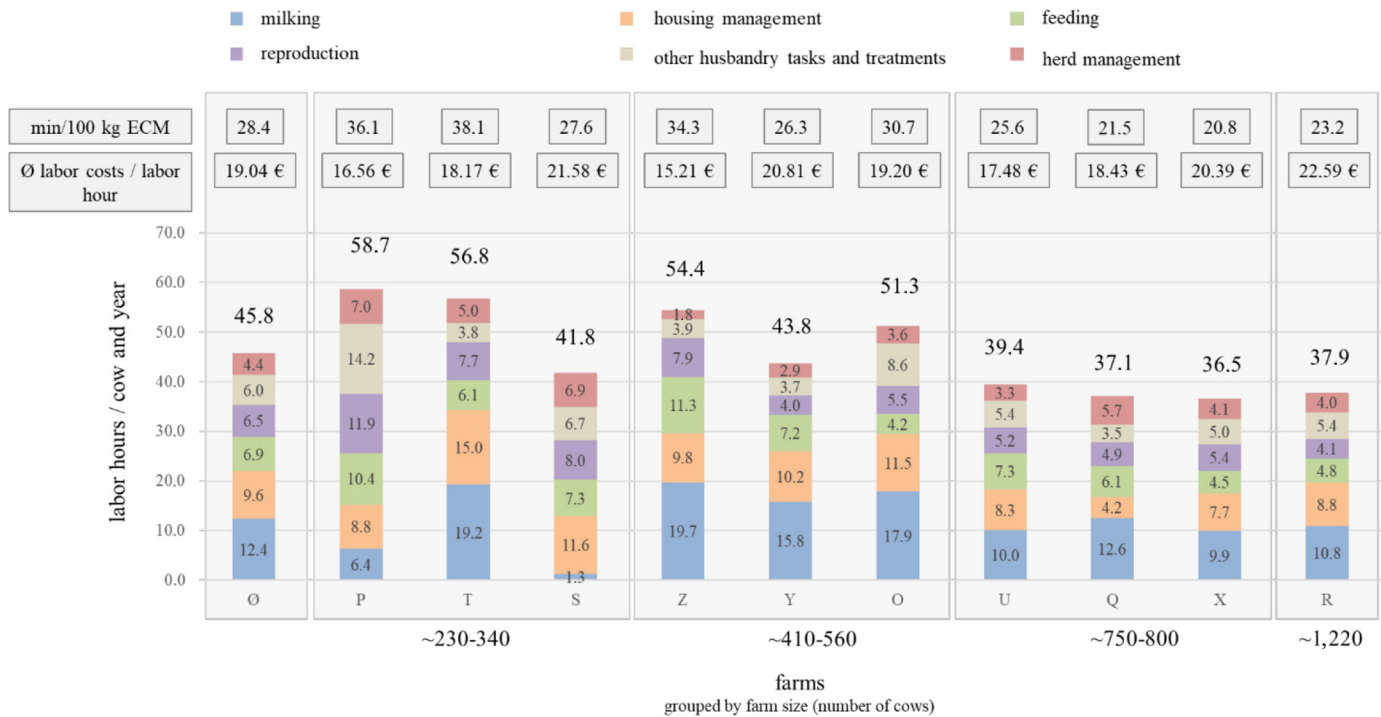


Figure 2. Labor hours per cow and year, labor efficiency (expressed as labor min/100 kg ECM), and labor costs/labor hour for the 10 project farms, sorted according to farm size. Ø = mean.

2018 and June 2022. The farm accountants were trained in 2 workshops and with personal assistance to record the respective data on a monthly basis. A project employee,

himself educated in accountancy, collected and validated the data on quarterly farm visits. For the analysis, a set of key figures was chosen, and their 12-mo rolling averages

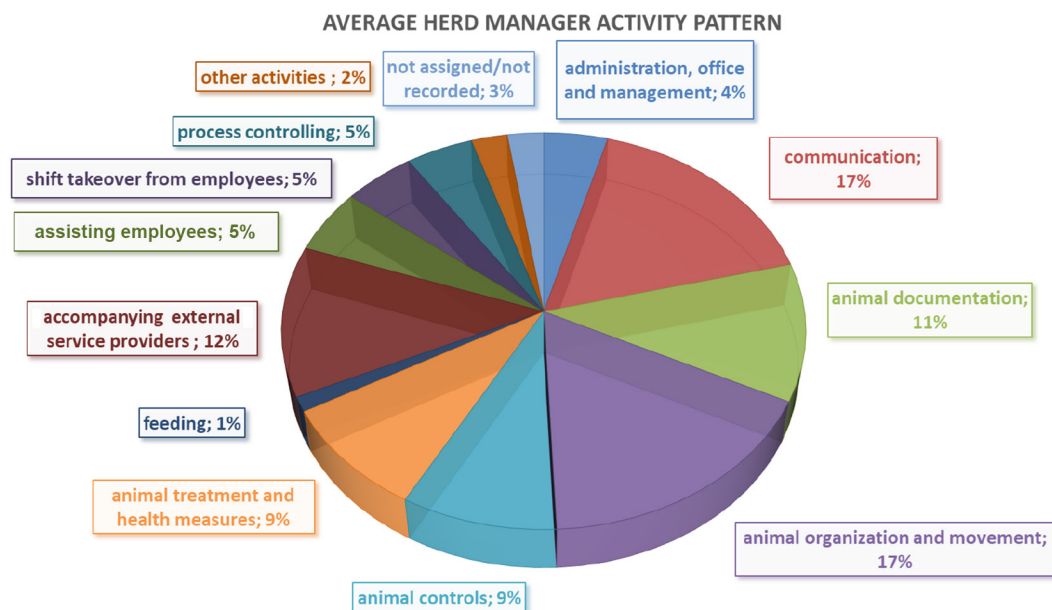


Figure 3. Average daily herd manager activity in the 10 project farms. A project employee documented the herd managers activity on a minute basis throughout a 3-wk period in each farm. A total of 1,092 h was documented and assigned to 13 main categories.

Table 3. Comparison of key figures (LSMeans) between 10 dairy farms with less (HM-Low) or more (HM-High) than average labor invested hours per cow and year for herd management

Key figure	Unit	HM-Low	HM-High	SE	P-value
Herd size	No. of cows	692	492	133	0.32
Labor h/cow per yr					
Total	h/cow/yr	45.36	46.18	4.11	0.89
Milking	h/cow/yr	14.86	9.87	2.52	0.20
Housing management	h/cow/yr	9.72	9.46	1.34	0.89
Feeding	h/cow/yr	6.97	6.88	1.13	0.96
Reproduction	h/cow/yr	5.33	7.59	1.01	0.15
Other husbandry tasks and treatments	h/cow/yr	5.37	6.64	1.52	0.57
Herd management (grouping criteria)	h/cow/yr	3.10	5.74	0.47	<0.01
Production and milk quality					
Marketed ECM per cow and year	kg	9,707	9,724	250	0.96
Test-day milk production	kg	32.1	32.2	1.0	0.91
Test-day herd SCC	K/mL	262	230	37	0.56
Culling, mortality, and replacements					
Culling rate	%	35.8	30.0	2.2	0.10
Mortality rate	%	8.0	7.6	1.3	0.82
Lifetime production	kg	31,052	36,046	2,406	0.18
Cow calvings	% of all calvings	67.6	71.3	2.7	0.36
Age at first calving	mo	24.8	24.7	0.5	0.93
Stillbirth rate	%	7.8	6.7	0.7	0.32
Loss during rearing period (3 mo)	%	6.4	5.3	2.0	0.69
Feeding					
Concentrate fed per kg milk sold	g/kg ECM	416	368	28	0.27
Average energy content of silages	MJ NEL	6.27	6.24	0.06	0.68
Average CP content of silages	%	12.1	11.8	0.4	0.70
Economics					
Carcass revenue	€	612	738	47	0.09
Animal health costs	€/100 kg ECM ¹	2.40	2.39	0.42	1.00
Variable costs	€/100 kg ECM ¹	38.10	38.26	1.70	0.95
Contribution margin	€/100 kg ECM ¹	-0.52	-0.15	1.43	0.86
Full costs	€/100 kg ECM ¹	44.37	45.50	2.15	0.72
Earnings before depreciation, interest, and taxes	€/100 kg ECM ¹	-3.08	-3.41	1.74	0.90

¹Shown as €/100 kg ECM sold.

created in June 2021 were used. The variables are listed in Tables 3 and 4 and included key figures from following areas: production level and milk quality (kilogram of marketed milk per cow and year, test-day milk production, and SCC), culling, mortality and replacement figures (culling rate, mortality rate [% euthanized and died], lifetime production of culled cows, % of cow calvings, age at first calving, stillbirth rate, and losses during rearing period [until 3 mo of age]), feeding (concentrate fed per kg milk sold [concentrate fed to youngstock included], average energy and protein content of silages [mean of all samples collected in 1-yr period], of each silage bunker a sample was collected during monthly visits [at least 5 samples per bunker] and analyzed using near infrared spectroscopy in the regional control laboratory [Landeskontrollverband Berlin-Brandenburg eV]), and economics including average carcass revenue per animal slaughtered, costs for animal health, variable costs, contribution margin, full costs and earnings before depreciation, interest, and taxes expressed as €/100 kg sold ECM ($\text{ECM} = \text{milk yield (kg/d)} \times \{[1.05 + (0.38 \times \text{fat \%}) + (0.21 \times \text{protein \%})]/3.28\}$) according to the Society of Nutrition Physiology (GfE, 2001).

Statistical Analysis

For analysis, the farms were grouped in

1. Farms investing more versus less than the average number of labor hours in herd management: herd management high (**HM-High**) versus herd management low (**HM-Low**).
2. Farms in which the herd manager spends more (**C-High**) versus less (**C-Low**) than average of his/her daily activities on controlling (defined as % of daily activity for animal controls, feeding, and process controlling, Table 2).

The groups were then analyzed for differences between the respective key figures using PROC MIXED in SAS 9.4 (SAS Institute, Cary, NC) with group as a fixed factor.

RESULTS

Labor Hours

The average invested labor hours per cow and year amounted to 45.8 across the 10 farms, with a minimum of

Table 4. Comparison of key figures (LSMeans) between 10 dairy farms with a herd manager showing less than average (C-L, n = 5) or more than average (C-H, n = 5) controlling activity (defined as % of daily activity for animal, feeding, and process control)

Key figure	Unit	C-Low	C-High	SE	P-value
Herd size	No. of cows	560	624	141	0.76
Daily activities					
Administration, office, and management	%	3.1	5.6	1.2	0.19
Communication	%	14.7	18.6	2.3	0.26
Animal documentation	%	10.2	12.3	2.4	0.55
Animal organization and movement	%	16.0	18.1	2.3	0.55
Animal controls	%	6.9	10.6	1.5	0.11
Animal measures and treatments	%	10.7	7.1	1.8	0.19
Feeding	%	0.5	1.9	0.4	0.06
Accompanying external service providers	%	15.3	9.2	3.1	0.22
Assisting employees	%	6.1	4.4	1.7	0.47
Shift takeover from employees	%	9.7	0	4.4	0.16
Process controlling	%	3.0	6.1	0.8	0.03
Other activities	%	2.3	2.6	0.6	0.72
Controlling activity (grouping criteria)	%	10.4	18.7	1.0	<0.01
Production and milk quality					
Marketed ECM per cow and year	kg	9,625	9,804	24	0.62
Test-day milk production	kg	31.4	32.9	0.91	0.26
Test-day herd SCC	K/mL	288	205	31	0.09
Culling, mortality, and replacements					
Culling rate	%	35.3	30.4	2.3	0.18
Mortality rate	%	9.7	5.9	0.9	0.02
Lifetime production	kg	30,400	36,698	2,207	0.08
Cow calvings	% of all calvings	65.8	73.2	2.2	0.04
Age at first calving	months	25.0	24.5	0.5	0.44
Stillbirth rate	%	8.0	6.5	0.7	0.16
Loss during rearing period (3 mo)	%	8.3	3.5	1.7	0.08
Feeding					
Concentrate fed per kg milk sold	g/kg ECM	420	363	27	0.18
Average energy content of silages	MJ NEL	6.28	6.23	0.06	0.60
Average CP content of silages	%	12.1	11.7	0.4	0.52
Economics ¹					
Carcass revenue	€	700	650	55	0.54
Animal health costs	€/100 kg ECM ¹	2.16	2.63	0.40	0.43
Variable costs	€/100 kg ECM ¹	38.05	38.31	1.70	0.92
Contribution margin	€/100 kg ECM ¹	-0.97	0.31	1.39	0.53
Full costs	€/100 kg ECM ¹	45.66	44.21	2.13	0.64
Earnings before depreciation, interest, and taxes	€/100 kg ECM ¹	-3.93	-2.56	1.71	0.59

¹Shown as €/100 kg ECM sold.

36.5 h and a maximum of 58.7 h and an average of €19.04 (€15.21–22.59) invested per labor hour (Figure 2). The labor efficiency expressed as invested minutes of labor to produce 100 kg of ECM averaged 28.4 (20.8–38.1). The lowest amount of labor hours invested and highest efficiency are observed in the 4 largest farms (U, Q, X, and R), with herd sizes between ~750 and 1,200 cows. The labor costs, which can also be interpreted as average salaries, show no interrelation with the herd size, labor hours per cow, or efficiency. The invested hours in herd management averaged 4.4 h/cow/yr, ranging between 1.8 and 7.0 h/cow/yr.

Interrelation Between Invested Labor Hours in Herd Management and Key Figures

The HM-Low farms (Z, Y, U, O, R) invested on average 3.10 h/cow/yr in the area of herd management, whereas

the HM-High (X, T, Q, S, P) farms invested 5.74 h/cow/yr on average, with no significant difference for any of the key figures observed (Table 3). For the culling rate ($P = 0.10$) and the carcass revenue ($P = 0.09$), a trend was observed with a 5.8% lower culling rate (35.8 vs. 30.0%) and an average of €126 lower revenue (€612 vs. 738) per carcass in the HC-low group.

Herd Manager Activities

The division of the documented 1,092 h of herd manager activity into the 12 main categories is depicted in Figure 3. On average, most time was invested in communication (9%–23%, min-max) and in animal organization and movement (11%–28%), followed by the accompanying of external service providers (e.g., veterinary, hoof trimmer, inseminator, 4%–29%) and in animal documentation (3%–19%). Overall, less than 10% of

the time each were used for animal controls (4%–16%) and animal treatment and (health) measures (2%–16%). Taken together, on average, another 10% of the time was spent on assisting employees (1%–11%) and taking over full shifts (0%–30%). The average herd manager spent the smallest part of his/her labor time in process controlling (0%–9%), administration, office, and management (1%–9%), feeding (0%–3%), and other activities outside of the dairy production (1%–5%). Only 3% (0%–11%) of the time was not assigned/not recorded.

Interrelation Between Controlling Activity of Herd Managers and Key Figures

In the C-High herds (O, R, X, T, P), the herd managers spent almost double the time on controlling activities (18.7% for animal controls, feeding, and process controlling, Table 4) compared with their counterparts (10.4% in C-Low herds, Z, Y, U, Q, S). In these herds, the SCC tended to be lower (288 thousand cells/mL vs. 205 thousand cells/mL, $P = 0.09$) and exhibited more favorable culling and replacement figures with a numerically lower culling rate (35.3% vs. 30.4%, $P = 0.18$), interrelating with a higher amount of cow calvings (65.8% vs. 73.2%, $P = 0.04$) and a higher lifetime production (30,400 kg vs. 36,698 kg, $P = 0.08$), as well as a markedly lower mortality rate (9.7% vs. 5.9%, $P = 0.02$). In these farms, furthermore, a trend in the loss during the rearing period during the first 3 mo of age (8.3% vs. 3.5%, $P = 0.08$) was observed. No significant differences or trends were observed in the key figures regarding feeding and economics.

DISCUSSION

In the current literature, few publications are available on the assessment of workflow on dairy farms (O'Donovan et al., 2008; Schick et al., 2009). Methods to describe labor input on farms categorized in different areas, also analyzing seasonal aspects, have been developed and widely applied (Cournut et al., 2018). Studies describing the invested hours of labor with a focus on herd management and its interrelation with farm performance are, however, scarce. Practical experience, however, shows that daily routines of herd managers differ between farms and are in some cases strongly led by the “demand of the day” principle. The aim of the presented case study was therefore to assess the hours invested in herd management, analyzing interrelations with the key figures for production, herd health, economics, and efficiency—thereby developing respectable protocols for future scientific analyses and practical assessments.

On average, the farms invested 45.8 h/cow/yr of labor. The farms sampled in this study are medium to large

sized in the international context (IFCN, 2022), and the documented hours of labor invested per cow and year are in line with different other studies. For 38 German and Swiss farms, Schick et al. (2009) describe a labor investment of 70 to 90 h/cow/yr in smallholder dairy farms (~40 cows), 60 to 70 h/cow/yr in medium-sized farms (~120 cows), and 50 to 60 h/cow/yr in large dairy farms (~1,000 cows). A study involving 16 Bavarian farms describes a total labor input between 30.1 and 74.7 h/cow/yr, with, however, a much higher input for milking compared with our study (21.5 h/cow/yr in conventional and 7.7 h/cow/yr in automated milking systems), most likely due to the difference in average parlor size and throughput (herd sizes: 57–235 cows; Mačuhová and Haidn, 2012). In a UK study, an average of 43.1 h/cow/yr in 228 dairy farms with an average size of 110 cows is reported (Wilson, 2011).

Our data analysis furthermore confirmed a higher efficiency of larger farms described by the key figures invested labor hours per cow and year (<40 h/cow/yr), as well as labor minutes to produce 100 kg of ECM (<25 min/100 kg of ECM; Hadley et al., 2002; Kompas and Che, 2006; Schick et al., 2009; Cournut et al., 2018; Pouloupoulou et al., 2018; Náglová and Rudinskaya, 2021). The analysis of amount of labor invested in the different areas of work shows a more similar distribution pattern in these 4 largest operations compared with the other 6 farms (Figure 2). A likely reason is their similar setup with a large central rotary milking parlor (40–60 units), short walking distances, deep-bedded cubicles or mattresses, and an external youngstock facility (Table 1). The smaller farms are more heterogeneous in their characteristics and therefore also the distribution of hours invested in the different areas. For example, farms P and S use single-unit robots for milking and thereby show only a small number of hours invested in milking. Farm P, however, additionally uses a herringbone parlor to milk fresh cows, decreasing the gain in efficiency by automatization. Farm T shows a high input in housing management (15.0 h/cow/yr) due to the daily cleaning and filling of the cubicles with straw bedding, and farm P has a high input in the area of reproduction due to the youngstock being distributed among multiple sites on the farm, not allowing efficient workflow. It, furthermore, needs to be highlighted that the invested hours of labor in a certain area do not per se speak for efficiency but can also be an expression of lack of labor force. For example, farm Q, with only 4.2 h/cow/yr invested in management of the housing system, the quality of the cubicle bedding material (filling grade and cleanliness) has repeatedly been evaluated as insufficient during the project period of several years. Therefore, such data and resulting recommendations always need to be combined with a sound analysis of the practical conditions and

control points of the housing, feeding, and animal health situation.

The number of hours invested in herd management with 4.4 h/cow/yr was 9.6% of the total labor hours invested. This is in line with an Irish study showing a proportion of 11% with 4.7 h/cow/yr (O'Donovan et al., 2008) and the proportion illustrated in Schick et al. (2009; no precise data available, only graphical illustration). Our analysis shows that in the study farms, the central tasks of a herd manager lie in organizational and animal healthcare duties involving communication and overall process control, animal controls and animal treatment and (health) measures, accompanying of external service providers, animal documentation, and administration and office tasks. However, a large heterogeneity between farms in the activities of their herd managers was observed. In some farms, focus on management and organizational tasks as named above is high (e.g., farm U and P, 82% and 77% of daily activities, average: 68%), whereas in others, herd managers are taking over many considered delegable activities for this farm size, such as animal organization and movement, assisting employees, or taking over full shifts (e.g., farms S and Z, 51% and 43%, average: 27%). The interviews showed that this can either be attributed to a deliberate choice to impose certain tasks on a herd manager due to organizational reasons or an absolute shortage of workforce.

It is furthermore noticeable that the 2 farms using robotic milking systems exhibit the highest labor input in respect to herd management (farm P: 7.0 h per cow and year, farm S: 6.9 h per cow and year; average of all farms: 4.4 h per cow and year). This could be attributed to possible higher labor input of the herd manager in the area of robot operation, as the responsibility for the reaction to alarms as well as handling of special needs animals in this area often lies in the area of the herd manager, also being the case in farms P and S in our dataset. Moreover, in line with the literature, in the robotic milking farm S, a noticeably lower total input of labor per cow and year was observed, compared with the other farms of a comparable size applying conventional milking systems (Figure 2, Latvala and Pyykkonen, 2005). This was not the case in the robotic milking farm P. The analysis of the workflow and structural conditions showed that this could be attributed to the lack of streamlined work processes due to the strong dispersion of the different facilities over the farm grounds.

We observed that in most of the farms, the controlling of the feeding process is largely disconnected from the person in charge of herd management, with only 1% (0%–3%) of activity on average invested in this area. The interviews revealed that in all farms somebody else at the management level of the farm (mostly farm manager) was responsible for implementing the control of

the feeding process and feed intake, whereas the ration formulation was in most farms in the hands of external consultants affiliated with a commercial feed company. This distribution of responsibilities implicates that the herd manager, being responsible for the control of animal health and productivity and thereby being the person with the most intensive cow contact on a daily basis, is not concomitantly responsible for the controlling and management of the feeding of the animals. As there is a very close connection between the animal health situation and feeding routines (Ingvarsen, 2006; Mulligan et al., 2006), this is to be seen as critical. The distribution of tasks in such a way implicates a high communicational effort with the risk of a possible loss of information and lag of action—all in all presenting a possible major risk factor for health and productivity.

Our analysis of different key figures in regard to the amount and quality of herd management work invested revealed a distinct differentiation between farms in their operational success when dividing between the amount of controlling activity performed by the herd managers. This underlines that it is not only per se the number of hours of labor invested in a certain area of work, but the quality of work that is delivered that is of decisive value. Herd managers spending more time in controlling activity achieve better results concerning the health and longevity of their herd, described by lower tank SCC, more favorable culling and replacement figures, and losses during the rearing period. These findings were irrespective of the herd size, in line with a recent study conducted in German dairy herds, showing that herd size has no influence on the welfare state of the herd but rather the time spent in animal care (Lindena and Hess, 2022). These results emphasize the importance of the human-animal relationship in animal husbandry, corresponding to studies showing a positive interrelation between the quality of this interaction and animal health status of the herd (Ivemeyer et al., 2011, 2018; Adler et al., 2019), highlighting the importance of the interrelation between herd-level and individual cow health management (Maltz, 2020).

It furthermore needs to be emphasized that these differences in animal health status between C-Low and C-High farms also implicate a difference in animal welfare (AHAW et al., 2023). Different studies describe the effect of the animal welfare status within a farm on the employee satisfaction and mental health (Kolstrup and Hultgren, 2011; Tremetsberger and Winckler, 2015). But also vice versa, studies have indicated an interrelation between the mental health status of farmers and animal welfare (Lunner Kolstrup et al., 2013; King et al., 2021). The effect of good process control should therefore be seen beyond the aspect of animal productivity, and future studies regarding work organization, as well as educa-

tional and governmental support programs, should lay their focus here as well.

Better milk quality and increased health status and longevity have a positive economic effect (Tremetsberger and Winckler, 2015; Hansen, 2023). In our analysis, this was not apparent when comparing the farms depending on their input of labor hours in herd management, as well as regarding the controlling activity of their herd managers. A likely explanation is that the economic success (described by the key figures: variable costs, contribution margin, full costs, and earnings before depreciation, interest, and taxes) of a farm is dependent on many different factors, for example, farm size and structure (available land and its quality), technological and labor efficiency, feed costs, and milk price (Gloy et al., 2002; Walsh et al., 2020). Analogously, no difference between costs for animal health was observed between the groups. After detailed analysis of the animal health management situation of the different farms throughout this project, we ascribe this to different strategies among farms: for example, some farms invest more in preventive medicine, achieving a good health status, whereas others have increased costs for animal health due to the need to invest more in curative medicine. Yet others have minimal expenses in this area owing to little necessity or because of failure to detect and treat diseased animals in time, leading to a higher culling rates.

Our results are in overall line with studies showing an interrelation between the farmers' or herd managers' attitudes toward organizational aspects and animal welfare (Kauppinen et al., 2013; Adler et al., 2019; Redfern et al., 2021). An aspect that should, in our eyes, be involved in future analogous studies is the relationship between the farm management and the veterinary, as different studies have highlighted the challenges and importance of communication and relationship with the herd veterinary for good herd health management (Kristensen and Enevoldsen, 2008; Derks et al., 2013, 2014; Svensson et al., 2018; Redfern et al., 2021; Ritter et al., 2021).

The "controlling activity" in relation to dairy herd management has rarely been reflected upon, and there is, to our knowledge, no definition to be found in the literature of what daily activities could be categorized as such. We therefore held an expert round discussion after the contemplation of the list of activities that we had observed during our study, considering literature from other industry processes (Jones et al., 2001; Vitezić and Lebefromm, 2019). This list is not exclusive, and depending on the production type, activities will be/can be added. We also encourage other researchers to critically reflect on this categorization and analyze effects of the quality of this work on the production outcome. Currently digital systems are vastly gaining in relevance in regard to controlling animal health and production

(de Oliveira et al., 2024). Aside from possible benefits of including them in animal health routines, the possible effect of such systems on production efficiency should be critically investigated. Labor time consumption due to false alarm rates and technique maintenance are often underestimated factors, having a negative effect on the efficacy of such routines (Penry, 2018).

In summary, our data highlight the importance of well-organized workflows and daily activities. These aspects need to be part of the designing process of facilities ahead of construction, as well as when renovations are planned, as they have a major effect on the effectiveness and thereby on the profitability of a farm. Our analysis shows that the aspects considered should go beyond the major routines on a dairy farm (milking, feeding, bedding management) and also involve the herd management. This is in line with a recent publication showing that increased organization in workflows decreases working hours of farmers and thereby may increase their quality of life (Hogan et al., 2023) and different articles describing that good work organization and leadership in the form of regular communication, planning meetings, and written task descriptions are of importance in employee satisfaction (Kolstrup, 2012; Thinius and Jakob, 2015). This is vastly increasing in importance as the structural change in the sector leads to increased size of dairy farms and the necessity of a hired workforce (Barkema et al., 2015), also highlighting the need for sound educational programs allowing for specialization with a special focus on organization and leadership (Hansen and Greve, 2015). These aspects are underlined by the large differences in the areas of responsibility as well as the daily activities of herd managers observed in our study, indicating that increased focus on work organization on the part of these farms could increase the animal health and welfare situation.

There is, however, a need to address a list of limitations of this study. The farms were selected and volunteered from a small pool of farms due to their excellent documentation. This may cause a selection bias, and the results are therefore only representative within this specific study population. Even though the farms were relatively similar in regard to their operational setup (conventional, confinement, all-year calving, German Holstein breed), differences in milking system, but also other aspects (youngstock rearing system, reproduction and health routines, and so on), may influence organizational aspects that cannot be accounted for in such a small sample size. Therefore, this investigation was considered a case study, with studies including many farms stratified for different aspects warranted in the future. An observational timeframe of 3 wk was chosen to allow a complete data compilation within one year. Even though these farms are all-year calving and by experience only minorly

influenced by seasonal aspects, the short observational timeframe does not allow for the capturing of possible seasonal variation. The study method of direct behavior documentation comprises a possible observer bias, as this lack of blinding could lead to overperformance or deviation from typical practices, thereby compromising the authenticity of the observations. The application of other more objective monitoring tools, such as cameras, has, however, to be seen as critical, or their application is not legally possible due to privacy concerns. The unobserved documentation by the persons themselves has, furthermore, the constraint of incorrect or biased documentation. With solid alternatives lacking, future studies should acknowledge the biases that come with the different methods and try to minimize this by applying precautionary measures (e.g., training the observing person in being discrete, blinding of the research question). Furthermore, as described above, future studies should analyze and discuss “the controlling activity” in the context of farm type and with a larger sample size, applying different analysis approaches. Aside from the detailed documentation of daily activities, an annual overview of the work intensity and tasks could be recorded and analyzed by, for example, the “Work Assessment Method” (Cournut et al., 2018). Future studies should enforce a more detailed description of the position “herd manager” and its tasks and responsibilities within different production systems and the interrelation with the work quality and animal production, health, welfare, and overall economic performance of dairy production systems.

CONCLUSIONS

This case study is, to our knowledge, the first describing the daily routines of herd managers not only assessed by interview but also by a detailed observation over many days and divided and analyzed on the level of different tasks. The prepared schemes and methods can be used for efficiency analysis on farms in practice and consulting, as well as for research purposes. Our data show a large heterogeneity between farms in the activities of their herd managers, and in some farms, original herd manager activities are postponed by taking over tasks that should be delegated. In most of the farms the controlling of the feeding process was largely disconnected from the person in charge of herd management, which implicates a possible risk factor for animal health. Herd managers who spent more time in controlling achieved better results concerning the health and longevity of their herd, presenting a positive effect on animal welfare. This analysis highlights the importance of sound work organization and process control as a basis for a productive and healthy herd. Given the limitations of the study (e.g., selection bias, observer bias, sample size), future

studies should assess the interrelation between workflow and labor efficiency, animal health status and employee satisfaction, and mental health in a larger sample size stratified for different farming systems using different analysis approaches.

NOTES

We thank the farms involved in this study for their trust and engagement. The study was conducted in the scope of a European Innovation Partnership (EIP)-agri project (“Die Entwicklung des KUH-mehr-WERT Navigators”). The publication of this manuscript was supported by the Open Access Publishing Fund of Leipzig University (Germany). All herd managers who participated in this study were informed about the purpose and procedures of the observation in advance and gave their voluntary consent to participate. The observations were conducted overtly, ensuring transparency and respect for personal autonomy. No sensitive personal data were collected. Outside the analyzing team, the data were anonymized to protect the identity of individuals and farms. The study protocol adhered to ethical guidelines for research involving human participants and did not require formal ethics committee approval under national regulations. The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: C-High = controlling activity high; C-Low = controlling activity low; EPI-AGRI = European Innovation Partnership for Agricultural Productivity and Sustainability; HM-High = herd management high; HM-Low = herd management low; min-max = minimum–maximum; PMR = partial mixed ration.

REFERENCES

- Adler, F., R. Christley, and A. Campe. 2019. Invited review: Examining farmers’ personalities and attitudes as possible risk factors for dairy cattle health, welfare, productivity, and farm management: A systematic scoping review. *J. Dairy Sci.* 102:3805–3824. <https://doi.org/10.3168/jds.2018-15037>.
- AHAW (EFSA Panel on Animal Health and Animal Welfare), S. S. Nielsen, J. Alvarez, D. J. Bicoût, P. Calistri, E. Canali, J. A. Drewe, B. Garin-Bastuji, J. L. Gonzales Rojas, C. Gortázar Schmidt, M. Herskin, V. Michel, M. Ángel Miranda Chueca, B. Padalino, H. Clare Roberts, H. Spoolder, K. Stahl, A. Velarde, A. Viltrop, A. De Boyer des Roches, M. Bak Jensen, J. Mee, M. Green, H.-H. Thulke, E. Bailly-Caumette, D. Candiani, E. Lima, Y. Van der Stede, and C. Winckler. 2023. Welfare of dairy cows. *EFSA J.* 21:e07993.
- Auernhammer, H. 1976. Eine integrierte Methode zur Arbeitszeitanalyse, Planzeiterstellung und Modellkalkulation landwirtschaftlicher Arbeiten, dargestellt an verschiedenen Arbeitsverfahren der Bullenmast. Technische Universität München.
- Barkema, H. W., M. A. G. von Keyserlingk, J. P. Kastelic, T. J. G. M. Lam, C. Luby, J.-P. Roy, S. J. LeBlanc, G. P. Keefe, and D. F. Kelton. 2015. Invited review: Changes in the dairy industry affecting dairy cattle health and welfare. *J. Dairy Sci.* 98:7426–7445. <https://doi.org/10.3168/jds.2015-9377>.

- Bewley, J., R. W. Palmer, and D. B. Jackson-Smith. 2001. modeling milk production and labor efficiency in modernized Wisconsin dairy herds. *J. Dairy Sci.* 84:705–716. [https://doi.org/10.3168/jds.S0022-0302\(01\)74525-0](https://doi.org/10.3168/jds.S0022-0302(01)74525-0).
- Brass, D. J. 1985. Technology and the structuring of jobs: Employee satisfaction, performance, and influence. *Organ. Behav. Hum. Decis. Process.* 35:216–240. [https://doi.org/10.1016/0749-5978\(85\)90036-6](https://doi.org/10.1016/0749-5978(85)90036-6).
- Britt, J. H., R. A. Cushman, C. D. Dechow, H. Dobson, P. Humblot, M. F. Hutjens, G. A. Jones, P. S. Ruegg, I. M. Sheldon, and J. S. Stevenson. 2018. Invited review: Learning from the future—A vision for dairy farms and cows in 2067. *J. Dairy Sci.* 101:3722–3741. <https://doi.org/10.3168/jds.2017-14025>.
- Cournut, S., S. Chauvat, P. Correa, J. C. D. Santos Filho, F. Diéguez, N. Hostiou, D. K. Pham, G. Servière, M. T. Sraïri, and A. Turlot. 2018. Analyzing work organization on livestock farm by the work assessment method. *Agron. Sustain. Dev.* 38:58. <https://doi.org/10.1007/s13593-018-0534-2>.
- de Kruif, A., R. Mansfeld, and M. Hoedemaker. 2006. Tierärztliche Bestandsbetreuung beim Milchrind. Georg Thieme Verlag.
- de Oliveira, F. M., G. A. S. Ferraz, A. L. G. André, L. S. Santana, T. Norton, and P. F. P. Ferraz. 2024. Digital and precision technologies in dairy cattle farming: A bibliometric analysis. *Animals (Basel)* 14:1832.
- Derks, M., H. Hogeveen, S. R. Kooistra, T. van Werven, and L. W. Tauer. 2014. Efficiency of dairy farms participating and not participating in veterinary herd health management programs. *Prev. Vet. Med.* 117:478–486.
- Derks, M., T. van Werven, H. Hogeveen, and W. D. J. Kremer. 2013. Veterinary herd health management programs on dairy farms in the Netherlands: Use, execution, and relations to farmer characteristics. *J. Dairy Sci.* 96:1623–1637.
- Edwards, J. P., B. Kuhn-Sherlock, B. T. D. Rue, and C. R. Eastwood. 2020. Technologies and milking practices that reduce hours of work and increase flexibility through milking efficiency in pasture-based dairy farm systems. *J. Dairy Sci.* 103:7172–7179.
- GfE. 2001. Empfehlungen zur Energie- und Nährstoffversorgung der Milchkühe und Aufzuchttrinder. DLG-Verlags GmbH, Frankfurt am Main.
- Gleeson, D., B. O'Brien, and K. O'Donovan. 2008. The labour input associated with calf care on Irish dairy farms. *Livest. Sci.* 116:82–89.
- Gloy, B. A., J. Hyde, and E. L. LaDue. 2002. Dairy farm management and long-term farm financial performance. *Agric. Resour. Econ. Rev.* 31:233–247.
- Green, M. J., and A. J. Bradley. 2012. Dairy Herd Health. CABI.
- Hadley, G., S. Harsh, and C. Wolf. 2002. Managerial and financial implications of major dairy farm expansions in Michigan and Wisconsin. *J. Dairy Sci.* 85:2053–2064.
- Hansen, B. G. 2023. Animal welfare and farm economy-exploring the relationship between dairy animal welfare indicators and economic efficiency. *Prev. Vet. Med.* 221:106058.
- Hansen, B. G., and A. Greve. 2015. The role of human and social capital in dairy farming. *Rural Soc.* 24:154–176.
- Hesse, A., S. Bertulat, and W. Heuwieser. 2017. Survey of work processes on German dairy farms. *J. Dairy Sci.* 100:6583–6591.
- Hogan, C., J. Kinsella, M. Beecher, and B. O'Brien. 2023. The impact of work organisation on the work life of people on pasture-based dairy farms. *Animal* 17:100686.
- Hogan, C., J. Kinsella, B. O'Brien, A. Markey, and M. Beecher. 2022. Estimating the effect of different work practices and technologies on labor efficiency within pasture-based dairy systems. *J. Dairy Sci.* 105:5109–5123.
- Hutt, G. 1993. Total quality and the dairy farm business organisation. Pages 130–135 in *Proc. Western Large Herd Management Conference*, Las Vegas, NV.
- Hutton, C. T., L. K. Fox, and D. D. Hancock. 1990. Mastitis control practices: Differences between herds with high and low milk somatic cell counts. *J. Dairy Sci.* 73:1135–1143.
- IFCN. 2022. International Farm Comparison Network - Dairy Report. ICFN Dairy.
- Ingvartsen, K. L. 2006. Feeding-and management-related diseases in the transition cow: Physiological adaptations around calving and strategies to reduce feeding-related diseases. *Anim. Feed Sci. Technol.* 126:175–213.
- Ivemeyer, S., U. Knierim, and S. Waiblinger. 2011. Effect of human-animal relationship and management on udder health in Swiss dairy herds. *J. Dairy Sci.* 94:5890–5902.
- Ivemeyer, S., C. Simantke, A. Ebinghaus, P. H. Poulsen, J. T. Sorensen, T. Rousing, R. Palme, and U. Knierim. 2018. Herd-level associations between human-animal relationship, management, fecal cortisol metabolites, and udder health of organic dairy cows. *J. Dairy Sci.* 101:7361–7374.
- Jones, A., Y. Yih, and E. Wallace. 2001. Monitoring and Controlling Operations. *The Handbook of Industrial Engineering*. John Wiley & Sons.
- Kaappinen, T., A. Valros, and K. M. Vesala. 2013. Attitudes of dairy farmers toward cow welfare in relation to housing, management and productivity. *Anthrozoos* 26:405–420.
- King, M. T. M., R. D. Matson, and T. J. DeVries. 2021. Connecting farmer mental health with cow health and welfare on dairy farms using robotic milking systems. *Anim. Welf.* 30:25–38.
- Kleiber, H. 1981. Verfahren der Milchproduktion. Landwirtschaftsverlag DDR, Berlin.
- Kolstrup, C. L. 2012. What factors attract and motivate dairy farm employees in their daily work? *Work* 41(Suppl 1):5311–5316.
- Kolstrup, C. L., and J. Hultgren. 2011. Perceived physical and psychosocial exposure and health symptoms of dairy farm staff and possible associations with dairy cow health. *J. Agric. Saf. Health* 17:111–125.
- Kompas, T., and T. N. Che. 2006. Technology choice and efficiency on Australian dairy farms. *Aust. J. Agric. Resour. Econ.* 50:65–83.
- König, F., A. Hancock, C. Wunderlich, M. Klawitter, T. Breuer, A. Simoni, K. Weimar, M. Drillich, and M. Iwersen. 2023. Systematic evaluation of different fresh cow monitoring procedures. *Animals (Basel)* 13:1231.
- König, S., and H. H. Swalve. 2024. Breeding for metabolic robustness and longevity in dairy cows. Pages 531–553 in *Production Diseases in Farm Animals: Pathophysiology, Prophylaxis and Health Management*. Springer.
- Kristensen, E., and C. Enevoldsen. 2008. A mixed methods inquiry: How dairy farmers perceive the value (s) of their involvement in an intensive dairy herd health management program. *Acta Vet. Scand.* 50:1–12.
- Latvala, T., and P. Pyykkonen. 2005. Profitability of and reasons for adopting automatic milking systems. XI Congress of the EAAE (European Association of Agricultural Economists), The Future of Rural Europe in the Global Agri-Food System. Copenhagen, Denmark.
- Lindena, T., and S. Hess. 2022. Is animal welfare better on smaller dairy farms? Evidence from 3,085 dairy farms in Germany. *J. Dairy Sci.* 105:8924–8945.
- Lunner Kolstrup, C., M. Kallioniemi, P. Lundqvist, H.-R. Kymäläinen, L. Stallones, and S. Brumby. 2013. International perspectives on psychosocial working conditions, mental health, and stress of dairy farm operators. *J. Agromed.* 18:244–255.
- Mačuhová, J., and B. Haidn. 2012. Labour input on Bavarian dairy farms with conventional or automatic milking. In *Proc. International Conference of Agricultural Engineering*, Valencia, Spain.
- Maltz, E. 2020. Individual dairy cow management: Achievements, obstacles and prospects. *J. Dairy Res.* 87:145–157.
- Mills, K. E., K. E. Koralesky, D. M. Weary, and M. A. G. von Keyserlingk. 2020. Dairy farmer advising in relation to the development of standard operating procedures. *J. Dairy Sci.* 103:11524–11534.
- Mothes, E., K. Zeihn, and J. Glanz. 1972. Technologie und Ökonomie einer Laufstallanlage für 1200 Milchkühe. *Deutsche Agrartechnik* 22(8).
- Mulligan, F. J., L. O'Grady, D. A. Rice, and M. L. Doherty. 2006. A herd health approach to dairy cow nutrition and production diseases of the transition cow. *Anim. Reprod. Sci.* 96:331–353.
- Náglóvá, Z., and T. Rudinskaya. 2021. Factors influencing technical efficiency in the EU dairy farms. *Agriculture* 11:1114.

- New Zealand Government. 2024. Job Profiles, Dairy Herd Manager. <https://www.careers.govt.nz/>.
- Noordhuizen, J. P. T. M. 2012. Dairy Herd Health and Management. A Guide for Veterinarians and Dairy Professionals. Mill Street, Packington, UK.
- O'Donovan, K., B. O'Brien, D. Ruane, J. Kinsella, and D. Gleeson. 2008. Labour input on Irish dairy farms and the effect of scale and seasonality. *J. Farm Manag.* 13:38–53.
- Penry, J. F. 2018. Mastitis control in automatic milking systems. *Vet. Clin. North Am. Food Anim.* 34:439–456.
- Pitkäranta, J., V. Kurkela, V. Huotari, M. Posio, and C. E. Halbach. 2019. Designing automated milking dairy facilities to maximize labor efficiency. *Vet. Clin. North Am. Food Anim.* 35:175–193.
- Poulopoulou, I., M. C. Nock, S. Steinmayer, C. Lambert, and M. Gauly. 2018. How can working time analysis contribute to the production efficiency of dairy farms in mountain regions? *Ital. J. Anim. Sci.* 17:489–498.
- Prendergast, R., M. D. Murphy, F. Buckley, and J. Upton. 2024. The effects of seasonality, management, infrastructure, and automation on the milking efficiency of herringbone and rotary milking parlors in Ireland. *J. Dairy Sci.* 107:917–932.
- Redfern, E. A., L. A. Sinclair, and P. A. Robinson. 2021. Dairy cow health and management in the transition period: The need to understand the human dimension. *Res. Vet. Sci.* 137:94–101.
- Reijers, H. A., I. Vanderfeesten, and W. M. P. van der Aalst. 2016. The effectiveness of workflow management systems: A longitudinal study. *Int. J. Inf. Manage.* 36:126–141.
- Ritter, C., L. Dorrestein, D. F. Kelton, and H. W. Barkema. 2021. Herd health and production management visits on Canadian dairy cattle farms: Structure, goals, and topics discussed. *J. Dairy Sci.* 104:7996–8008.
- Rodriguez, L. G. 2021. Assessing dairy farm employees health. Master thesis. South Dakota State University, Brookings, SD.
- Sanftleben, P., U. Knierim, H.-J. Herrmann, C. Mueller, and E. Von Borell. 2007. Critical control points (CCP) in dairy housing and management. *Zuchtungskunde* 79:339–362.
- Sauer, J., and U. Latacz-Lohmann. 2015. Investment, technical change and efficiency: Empirical evidence from German dairy production. *Eur. Rev. Agric. Econ.* 42:151–175.
- Schären-Bannert, M., P. Hufe, B. Waurich, A. Wöckel, W. Wippermann, E. Bannert, S. Hilke, G. Hermenau, J. Wittich, C. Felgentreu, F. Fröhlich, and D. May. 2023. Abschlussbericht des EIP-Projektes “Die Entwicklung des KUH-mehr-WERT Navigators.” Groß-Kreutz.
- Schick, M., K. Heitkämper, and M. Riegel. 2009. Work economics and ergonomics in dairy farming. In *Proc. International Commission of Agricultural and Biological Engineers, Section V. Conference “Technology and Management to Increase the Efficiency in Sustainable Agricultural Systems,”* Rosario, Argentina. ART-Schriftenreihe, No. 3, 52–59 ref. 8.
- Svensson, C., K. Alvåsen, A. C. Eldh, J. Frössling, and H. Lomander. 2018. Veterinary herd health management—Experience among farmers and farm managers in Swedish dairy production. *Prev. Vet. Med.* 155:45–52.
- Thinius, M., and M. C. Jakob. 2015. A strategy for workplace health promotion on German dairy farms. *Agric. Eng. Int. CIGR J.* 17:173–180.
- Tremetsberger, L., and C. Winckler. 2015. Effectiveness of animal health and welfare planning in dairy herds: A review. *Anim. Welf.* 24:55–67.
- Vitezić, N., and U. Lebefromm. 2019. Production Controlling in the Digital Age. University of Rijeka, Faculty of Economics and Business.
- Walsh, J., R. Parsons, Q. Wang, and D. Conner. 2020. What makes an organic dairy farm profitable in the United States? Evidence from 10 years of farm level data in Vermont. *Agriculture* 10:17.
- Wilson, P. 2011. Decomposing variation in dairy profitability: The impact of output, inputs, prices, labour and management. *J. Agric. Sci.* 149:507–517.
- Yeamkong, S., S. Koonawootrittriron, M. A. Elzo, and T. Suwanasopee. 2010. Effect of experience, education, record keeping, labor and decision making on monthly milk yield and revenue of dairy farms supported by a private organization in Central Thailand. *Asian-Australas. J. Anim. Sci.* 23:814–824.